

TUGAS PEKAN 8
PEMROGRAMAN ALGORITMA DAN PEMROGRAMAN

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TUGAS

MEMBUAT OPERASI ASSIGMENT DENGAN GRAPHICAL USER INTERFACE (GUI) PADA JAVA

1. KODE PROGRAM

```
1 package pekan8_2511533004;
2
3
4 import java.awt.EventQueue;
5
6 import javax.swing.JFrame;
7 import javax.swing.JPanel;
8 import javax.swing.border.EmptyBorder;
9 import java.awt.Color;
10 import javax.swing.JLabel;
11 import javax.swing.JOptionPane;
12
13 import java.awt.Font;
14 import javax.swing.SwingConstants;
15 import javax.swing.JTextField;
16 import javax.swing.JButton;
17 import java.awt.event.ActionListener;
18 import java.awt.event.ActionEvent;
```

Code program ini digunakan sebagai Import library dari Java untuk membuat aplikasi GUI. Lalu dapat dilanjutkan dengan menginisiasikan variabel variable dan membuat pesan peringatan dan pesan error.

```
20 public class AlproPekan8_2511533004 extends JFrame {
21
22     private static final long serialVersionUID = 1L;
23     private JPanel contentPane;
24     private JTextField tfBill1;
25     private JTextField tfBill2;
26     private JTextField txtPenjumlahan;
27     private JTextField txtPengurangan;
28     private JTextField txtPerkalian;
29     private JTextField txtPembagian;
30     private JTextField txtMod;
31     private void pesanPeringatan(String pesan) {
32         JOptionPane.showMessageDialog(this, pesan, "Peringatan", JOptionPane.WARNING_MESSAGE);
33     }
34     private void pesanError(String pesan) {
35         JOptionPane.showMessageDialog(this, pesan, "Kesalahan", JOptionPane.ERROR_MESSAGE);
36     }
37 }
```

Lalu dapat menambahkan method main

```
41 public static void main(String[] args) {
42     EventQueue.invokeLater(new Runnable() {
43         public void run() {
44             try {
45                 AlproPekan8_2511533004 frame = new AlproPekan8_2511533004();
46                 frame.setVisible(true);
47             } catch (Exception e) {
48                 e.printStackTrace();
49             }
50         }
51     });
52 }
```

Setelah itu lanjutkan dengan mengatur size, warna background, dan layout sesuai yang di inginkan.

```
57 public AlproPekan8_2511533004() {
58     setTitle("Operator Assignment");
59     setBackground(new Color(240, 240, 240));
60     setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
61     setBounds(100, 100, 450, 600);
62     contentPane = new JPanel();
63     contentPane.setBackground(Color.PINK);
64     contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
65     setContentPane(contentPane);
66     contentPane.setLayout(null);
67
68     JLabel Bil1 = new JLabel("Bilangan 1");
69     Bil1.setForeground(Color.BLACK);
70     Bil1.setFont(new Font("Times New Roman", Font.BOLD, 12));
71     Bil1.setBounds(33, 64, 80, 13);
72     contentPane.add(Bil1);
73
74     JLabel lblTitle = new JLabel("OPERATOR ASSIGNMENT");
75     lblTitle.setBounds(113, 10, 219, 22);
76     lblTitle.setForeground(Color.BLACK);
77     lblTitle.setHorizontalAlignment(SwingConstants.CENTER);
78     lblTitle.setFont(new Font("Times New Roman", Font.BOLD, 18));
79     contentPane.add(lblTitle);
80
81     JLabel Bil2 = new JLabel("Bilangan 2");
82     Bil2.setForeground(Color.BLACK);
83     Bil2.setFont(new Font("Times New Roman", Font.BOLD, 12));
84     Bil2.setBounds(33, 97, 80, 13);
85     contentPane.add(Bil2);
86
87     tfBil1 = new JTextField();
88     tfBil1.setHorizontalAlignment(SwingConstants.CENTER);
89     tfBil1.setFont(new Font("Times New Roman", Font.BOLD, 12));
90     tfBil1.setBounds(102, 61, 96, 19);
91     contentPane.add(tfBil1);
92     tfBil1.setColumns(10);
93
94     tfBil2 = new JTextField();
95     tfBil2.setHorizontalAlignment(SwingConstants.CENTER);
96     tfBil2.setFont(new Font("Times New Roman", Font.BOLD, 12));
97     tfBil2.setColumns(10);
98     tfBil2.setBounds(102, 94, 96, 19);
99     contentPane.add(tfBil2);
100
101     JLabel lblNewLabel = new JLabel("HASIL OPERASI");
102     lblNewLabel.setForeground(Color.BLACK);
103     lblNewLabel.setFont(new Font("Times New Roman", Font.BOLD, 12));
104     lblNewLabel.setBounds(33, 158, 96, 13);
105     contentPane.add(lblNewLabel);
106
107     JLabel lblNewLabel_1 = new JLabel("Assignment Penjumlahan");
108     lblNewLabel_1.setForeground(Color.BLACK);
109     lblNewLabel_1.setFont(new Font("Times New Roman", Font.PLAIN, 12));
110     lblNewLabel_1.setBounds(35, 200, 180, 13);
111     contentPane.add(lblNewLabel_1);
112
113     JLabel lblNewLabel_1_1_1 = new JLabel("Assignment Pengurangan");
114     lblNewLabel_1_1_1.setForeground(Color.BLACK);
115     lblNewLabel_1_1_1.setFont(new Font("Times New Roman", Font.PLAIN, 12));
116     lblNewLabel_1_1_1.setBounds(35, 270, 180, 13);
117     contentPane.add(lblNewLabel_1_1_1);
118
119     JLabel lblNewLabel_1_1_2 = new JLabel("Assignment Hasil Sisa Bagi");
120     lblNewLabel_1_1_2.setForeground(Color.BLACK);
121     lblNewLabel_1_1_2.setFont(new Font("Times New Roman", Font.PLAIN, 12));
122     lblNewLabel_1_1_2.setBounds(33, 480, 180, 13);
123     contentPane.add(lblNewLabel_1_1_2);
124
125     JLabel lblNewLabel_1_1_3 = new JLabel("Assignment Perkalian");
126     lblNewLabel_1_1_3.setForeground(Color.BLACK);
127     lblNewLabel_1_1_3.setFont(new Font("Times New Roman", Font.PLAIN, 12));
128     lblNewLabel_1_1_3.setBounds(33, 340, 180, 13);
129     contentPane.add(lblNewLabel_1_1_3);
130
131     JLabel lblNewLabel_1_1_3_1 = new JLabel("Assignment Pembagian");
132     lblNewLabel_1_1_3_1.setForeground(Color.BLACK);
133     lblNewLabel_1_1_3_1.setFont(new Font("Times New Roman", Font.PLAIN, 12));
134     lblNewLabel_1_1_3_1.setBounds(33, 410, 180, 13);
135     contentPane.add(lblNewLabel_1_1_3_1);
136
137     JLabel lblNewLabel_2 = new JLabel("Penjumlahan");
138     lblNewLabel_2.setForeground(Color.BLACK);
139     lblNewLabel_2.setHorizontalAlignment(SwingConstants.LEFT);
140     lblNewLabel_2.setFont(new Font("Times New Roman", Font.BOLD, 12));
141     lblNewLabel_2.setBounds(33, 225, 80, 13);
142     contentPane.add(lblNewLabel_2);
143 }
```

```

144 JLabel lblNewLabel_2_1 = new JLabel("Pengurangan");
145 lblNewLabel_2_1.setForeground(Color.BLACK);
146 lblNewLabel_2_1.setHorizontalAlignment(SwingConstants.LEFT);
147 lblNewLabel_2_1.setFont(new Font("Times New Roman", Font.BOLD, 12));
148 lblNewLabel_2_1.setBounds(33, 290, 80, 13);
149 contentPane.add(lblNewLabel_2_1);
150
151 JLabel lblNewLabel_2_1_1 = new JLabel("Modulo");
152 lblNewLabel_2_1_1.setForeground(Color.BLACK);
153 lblNewLabel_2_1_1.setHorizontalAlignment(SwingConstants.LEFT);
154 lblNewLabel_2_1_1.setFont(new Font("Times New Roman", Font.BOLD, 12));
155 lblNewLabel_2_1_1.setBounds(33, 500, 80, 13);
156 contentPane.add(lblNewLabel_2_1_1);
157
158 JLabel lblNewLabel_2_1_2 = new JLabel("Perkalian");
159 lblNewLabel_2_1_2.setForeground(Color.BLACK);
160 lblNewLabel_2_1_2.setHorizontalAlignment(SwingConstants.LEFT);
161 lblNewLabel_2_1_2.setFont(new Font("Times New Roman", Font.BOLD, 12));
162 lblNewLabel_2_1_2.setBounds(33, 360, 80, 13);
163 contentPane.add(lblNewLabel_2_1_2);
164
165 JLabel lblNewLabel_2_1_2_1 = new JLabel("Pembagian");
166 lblNewLabel_2_1_2_1.setForeground(Color.BLACK);
167 lblNewLabel_2_1_2_1.setHorizontalAlignment(SwingConstants.LEFT);
168 lblNewLabel_2_1_2_1.setFont(new Font("Times New Roman", Font.BOLD, 12));
169 lblNewLabel_2_1_2_1.setBounds(33, 430, 80, 13);
170 contentPane.add(lblNewLabel_2_1_2_1);
171
172 txtPenjumlahan = new JTextField();
173 txtPenjumlahan.setHorizontalAlignment(SwingConstants.CENTER);
174 txtPenjumlahan.setFont(new Font("Times New Roman", Font.BOLD, 12));
175 txtPenjumlahan.setBounds(133, 222, 60, 20);
176 contentPane.add(txtPenjumlahan);
177 txtPenjumlahan.setColumns(10);
178
179 txtPengurangan = new JTextField();
180 txtPengurangan.setHorizontalAlignment(SwingConstants.CENTER);
181 txtPengurangan.setFont(new Font("Times New Roman", Font.BOLD, 12));
182 txtPengurangan.setColumns(10);
183 txtPengurangan.setBounds(133, 287, 60, 20);
184 contentPane.add(txtPengurangan);
185
186 txtPerkalian = new JTextField();
187 txtPerkalian.setHorizontalAlignment(SwingConstants.CENTER);
188 txtPerkalian.setFont(new Font("Times New Roman", Font.BOLD, 12));
189 txtPerkalian.setColumns(10);
190 txtPerkalian.setBounds(133, 357, 60, 20);
191 contentPane.add(txtPerkalian);
192
193 txtPembagian = new JTextField();
194 txtPembagian.setHorizontalAlignment(SwingConstants.CENTER);
195 txtPembagian.setFont(new Font("Times New Roman", Font.BOLD, 12));
196 txtPembagian.setColumns(10);
197 txtPembagian.setBounds(133, 427, 60, 20);
198 contentPane.add(txtPembagian);
199
200 txtMod = new JTextField();
201 txtMod.setHorizontalAlignment(SwingConstants.CENTER);
202
203 txtMod.setFont(new Font("Times New Roman", Font.BOLD, 12));
204 txtMod.setColumns(10);
205 txtMod.setBounds(133, 497, 60, 20);
206 contentPane.add(txtMod);
207
208 JButton btnProses = new JButton("Proses");
209 btnProses.setForeground(Color.BLACK);
210 btnProses.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {

```

Setelah mengatur size, font, dan background dan layoutnya, dapat melanjutkan dengan menginputkan Syntax yang berisikan Operator di operasikan.

```
212●         if (tfBil1.getText().trim().isEmpty()) {
213             pesanPeringatan("Bilangan 1 harus diisi!");
214             return;
215         }
216●         if (tfBil2.getText().trim().isEmpty()) {
217             pesanPeringatan("Bilangan 2 harus diisi");
218             return;
219         }
220
221●         try {
222             int a = Integer.parseInt(tfBil1.getText());
223             int b = Integer.parseInt(tfBil2.getText());
224
225             //Operasi
226             txtPenjumlahan.setText(String.valueOf(a + b));
227             txtPengurangan.setText(String.valueOf(a - b));
228             txtPerkalian.setText(String.valueOf(a * b));
229
230●             if (b == 0) {
231                 txtPembagian.setText("Error");
232                 txtMod.setText("Error");
233                 pesanError("Bilangan 2 tidak boleh 0!");
234●             } else {
235                 txtPembagian.setText(String.valueOf(a/b));
236                 txtMod.setText(String.valueOf(a%b));
237             }
238●         } catch (NumberFormatException ex) {
239             pesanError("Bilangan 1 dan 2 harus angka!");
```

Hasil Outputnya

The image shows two side-by-side screenshots of a Java Swing application titled "Operator Assignment". The application has a pink background and a title bar with standard window controls.

Left Screenshot (Input State):

- OPERATOR ASSIGNMENT** (Title)
- Bilangan 1**: Input field with value 31.
- Bilangan 2**: Input field with value 1.
- Proses**: Button.
- HASIL OPERASI** (Section Header)
- Assignment Penjumlahan**: Penjumlahan input field with value 32.
- Assignment Pengurangan**: Pengurangan input field with value 30.
- Assignment Perkalian**: Perkalian input field with value 31.
- Assignment Pembagian**: Pembagian input field with value 31.
- Assignment Hasil Sisa Bagi**: Modulo input field with value 0.

Right Screenshot (Output State):

- OPERATOR ASSIGNMENT** (Title)
- Bilangan 1**: Input field with value 31.
- Bilangan 2**: Input field with value 1.
- Proses**: Button.
- HASIL OPERASI** (Section Header)
- Assignment Penjumlahan**: Penjumlahan input field with value 32.
- Assignment Pengurangan**: Pengurangan input field with value 30.
- Assignment Perkalian**: Perkalian input field with value 31.
- Assignment Pembagian**: Pembagian input field with value 31.
- Assignment Hasil Sisa Bagi**: Modulo input field with value 0.

Program ini mensimulasikan Operator Assignment yang di buat dan di design dari GUI

2. BAHASA NATURAL

1. Mulai
2. Program menampilkan form dengan dua kotak input:
3. nisisasi Bil 1, Bil 2, Hasil Penjumlahan, Hasil Pengurangan, Hasil Pembagian, Hasil Perkalian, Hasil Modulus
4. Input Bil 1, Bil 2
5. menekan tombol “Proses”.
6. Program mengambil bil 1 dan bil 2 dari input.
7. Program melakukan operasi berikut:
 - Menghitung penjumlahan ($X + Y$)
 - Menghitung pengurangan ($X - Y$)
 - Menghitung perkalian ($X \times Y$)
 - Menghitung pembagian ($X \div Y$)
 - Menghitung modulo atau sisa bagi ($X \% Y$)
8. Program menampilkan hasil setiap operasi pada kotak hasil masing-masing:
9. Selesai

3. PSEUDOCODE

JUDUL:

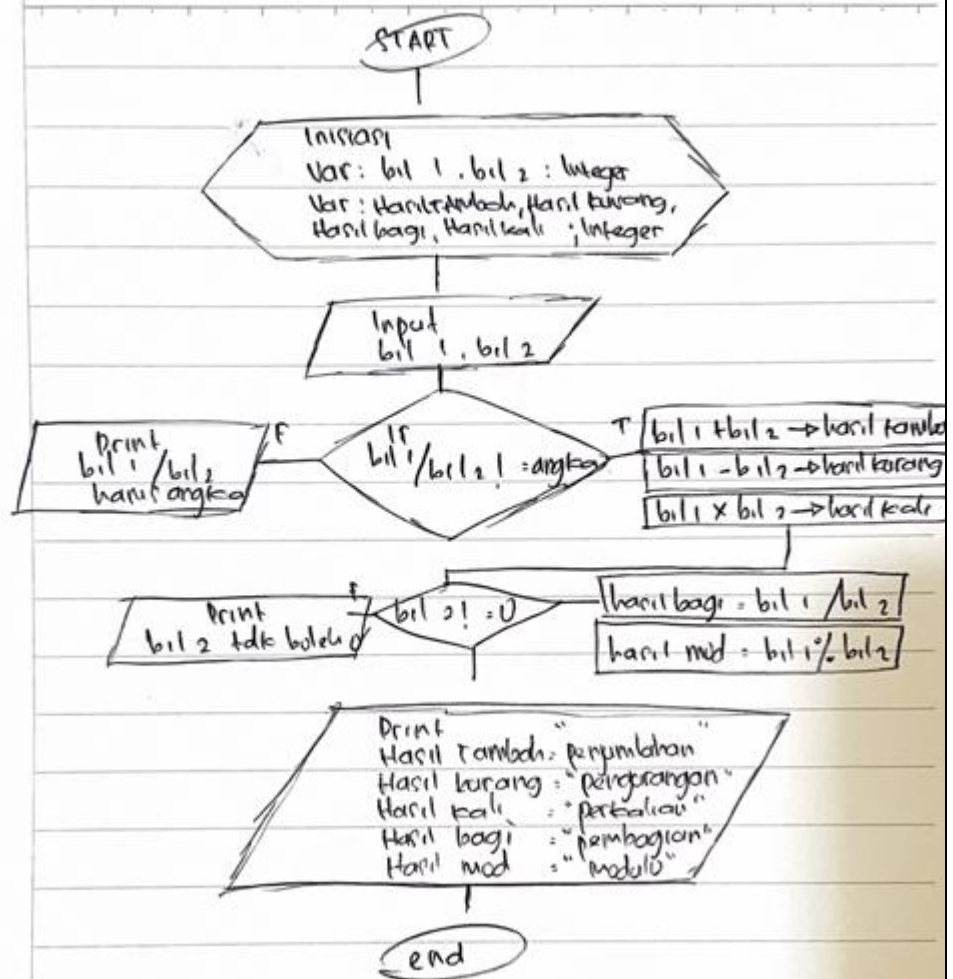
Program Operator Assigment

Program digunakan untuk menampilkan hasil perhitungan aritmataika (penjumlahan, pengurangan, perkalian, pembagian, dan modulo)

DEKLARASI:

Var = bil1, bil2 : Integer

Var = hasilTambah, hasilKurang, hasilKali, hasilBagi, hasilMod: Integer



4. FLOWCHART

